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United States Patent	12414013
Kind Code	B2
Date of Patent	September 09, 2025
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Apparatus and method for resource allocation in wireless communication system

Abstract

An electronic device of the disclosure may comprise: a communication circuit and at least one processor, wherein the at least one processor is configured to: control the communication circuit to transmit first capability information of the electronic device to a base station, identify, based on data throughput, whether an additional resource is needed, and based on identifying that the additional resource is needed, control the communication circuit to transmit an additional resource request message to the base station, wherein the first capability information comprises a request related to a basic resource, and the basic resource comprises at least one of a number of component carriers (CCs), a number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band.

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Appl. No.: 17/901730

Filed: September 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220417795 A1	Dec. 29, 2022

Foreign Application Priority Data

KR	10-2021-0034788	Mar. 17, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/KR2022/003707 20220317 PENDING child-doc US 17901730

Publication Classification

Int. Cl.: H04W28/20 (20090101); H04W8/24 (20090101)

U.S. Cl.:

CPC H04W28/20 (20130101); H04W8/24 (20130101);

Field of Classification Search

CPC: H04W (28/20); H04W (8/24); H04W (60/04); H04W (72/0453); H04W (72/21); H04W (72/52); H04W (72/51); H04W (88/02); H04B (7/0413); H04L (5/001); H04L (5/0064)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/KR2022/003707 designating the United States, filed on Mar. 17, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0034788, filed on Mar. 17, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

Field

(1) The disclosure relates generally to a wireless communication system and, for example, to an apparatus and a method for resource allocation in a wireless communication system.

Description of Related Art

(2) According to an increase in communication speed, terminal power consumption is increased, and thus overtemperature has become a problem. In order to reduce power and heating, terminals control a particular function, based on an internal thermistor or a power consumption amount.

SUMMARY

(3) Embodiments of the disclosure provide an apparatus and a method for controlling resource allocation by a terminal in a wireless communication system.

(4) Embodiments of the disclosure provide an apparatus and a method for being allocated a minimum resource by a terminal when the terminal accesses a network in a wireless communication system.

(5) Embodiments of the disclosure provide an apparatus and a method for being allocated an additional resource by a terminal, based on data throughput in a wireless communication system.

(6) Embodiments of the disclosure provide an apparatus and a method for identifying a minimum number of antenna elements by a terminal, based on received signal strength in a wireless communication system.

(7) An electronic device in a wireless communication system according to an example embodiment of the disclosure may include: a communication circuit and at least one processor operatively connected to the communication circuit, wherein the at least one processor is configured to: control the communication circuit to transmit first capability information of the electronic device to a base station; identify, based on data throughput, whether an additional resource is needed; and based on identifying that the additional resource is needed, control the communication circuit to transmit an additional resource request message to the base station, the first capability information

corresponding to a request related to a basic resource, and the basic resource being related to at least one of the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band.

(8) An electronic device in a wireless communication system according to an example embodiment of the disclosure may include: a communication circuit and at least one processor operatively connected to the communication circuit, wherein the at least one processor is configured to: control the communication circuit to transmit first capability information of the electronic device to a base station; control the communication circuit to transmit an overheating assistance message to the base station; identify, based on data throughput, whether an additional resource is needed; and based on identifying that the additional resource is needed, control the communication circuit to transmit an additional resource request message to the base station, the overheating assistance message including information on a basic resource, and the information on the basic resource including at least one of information on the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth.

(9) A method performed by an electronic device in a wireless communication system according to an example embodiment of the disclosure may include: transmitting first capability information of the electronic device to a base station; identifying, based on data throughput, whether an additional resource is needed; and based on identifying that the additional resource is needed, transmitting an additional resource request message to the base station, wherein the first capability information corresponds to a request related to a basic resource, and the basic resource is related to at least one of the number of component carriers (CCs), the number of multiple-input multiple output (MIMO) layers, or a bandwidth in an operating band.

(10) A method performed by an electronic device in a wireless communication system according to an example embodiment of the disclosure may include: transmitting first capability information of the electronic device to a base station; transmitting an overheating assistance message to the base station; identifying, based on data throughput, whether an additional resource is needed; and based on identifying that the additional resource is needed, transmitting an additional resource request message to the base station, wherein the overheating assistance message includes information on a basic resource, and the information on the basic resource includes at least one of information on the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth.

(11) According to an apparatus and a method of various example embodiments of the disclosure, the overheating of a device may be prevented or reduced by controlling resource allocation.

(12) In addition, according to an apparatus and a method of various example embodiments of the disclosure, the power consumption of a device may be optimized by controlling resource allocation.

(13) In addition, according to an apparatus and a method of various example embodiments of the disclosure, the time for maintenance of 5th generation (5G) communication connection may be maximized or increased.

(14) In addition, according to an apparatus and a method of various example embodiments of the disclosure, long term evolution (LTE) fallback may be prevented or reduced.

(15) In addition, according to an apparatus and a method of various example embodiments of the disclosure, the overheating of a device may be prevented or reduced in a 5G standalone (SA) mode.

(16) In addition, according to an apparatus and a method of various example embodiments of the disclosure, resource allocation may be efficiently performed.

(17) In addition, according to an apparatus and a method of various example embodiments of the disclosure, a terminal may be managed to operate at an operation securing temperature or lower.

(18) Effects which may be acquired by the disclosure are not limited to the effects described above, and other effects that have not been mentioned may be clearly understood by those skilled in the art from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a block diagram illustrating an example electronic device in a network environment according to various embodiments;
- (3) FIG. 2 is a diagram illustrating an example of a wireless communication system according to various embodiments;
- (4) FIG. 3 is a signal flow diagram illustrating an example operation of signaling for resource allocation according to various embodiments;
- (5) FIG. 4 is a signal flow diagram illustrating an example operation of signaling for resource allocation according to various embodiments;
- (6) FIG. 5 is a signal flow diagram illustrating an example operation of signaling for identifying the number of antenna elements according to various embodiments;
- (7) FIG. 6 is a flowchart illustrating an example operation of a terminal for configuring information on an additional resource according to various embodiments;
- (8) FIG. 7 is a flowchart illustrating an example operation of a base station for resource allocation according to various embodiments; and
- (9) FIG. 8 is a flowchart illustrating an example operation of a terminal for resource allocation according to various embodiments.

DETAILED DESCRIPTION

- (10) The terms used in the disclosure are merely for the purpose of describing various embodiments and are not intended to limit the scope of the embodiments. A singular expression may include a plural expression unless they are definitely different in a context. All terms used herein, including technical and scientific terms, may have the same meaning as those commonly understood by a person skilled in the art to which the disclosure pertains. Such terms as those defined in a generally used dictionary may be interpreted to have the meanings equal or similar to the contextual meanings in the relevant field of art, and are not to be interpreted to have ideal or excessively formal meanings unless clearly defined in the disclosure. In some cases, even the term defined in the disclosure should not be interpreted to exclude embodiments of the disclosure.
- (11) In addition, in the disclosure, although the expression “more than” or “less than” is used in order to determine whether or not a particular condition is fulfilled or satisfied, this is only an example and does not exclude the expression “equal to or more than” or “equal to or less than”. The expression “equal to or more than” may be replaced with “more than”, the expression “equal to or less than” may be replaced with “less than”, and the expression “equal to or more than and less than” may be replaced with “more than and equal to or less than” in the conditions above.
- (12) A metric for signal gain, signal quality, etc. used in the disclosure may include one of, for example, and without limitation, beam reference signal received power (BRSRP), reference signal received power (RSRP), reference signal received quality (RSRQ), received signal strength indicator (RSSI), signal to interference and noise ratio (SINR), carrier to interference and noise ratio (CINR), error vector magnitude (EVM), bit error rate (BER), and block error rate (BLER). In addition to the described examples, other terms having the same or similar technical meaning as the examples or other indexes indicating signal quality may be also used.
- (13) In addition, the disclosure includes terms used in some communication protocols (e.g., new radio (NR) or long term evolution (LTE) defined in 3.sup.rd generation partnership project (3GPP)) to explain various embodiments, but the terms merely correspond to examples. Various embodiments of the disclosure may also be easily modified and then applied to other

communication systems.

(14) The disclosure relates generally to an apparatus and a method for resource allocation in a wireless communication system. For example, in the disclosure, a technology of initiatively/pre-emptively controlling resource allocation by a terminal to prevent and/or reduce the overheating of the terminal will be described.

(15) FIG. 1 is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(16) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(17) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised

learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(18) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(19) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(20) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(21) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(22) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(23) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(24) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(25) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(26) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(27) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a

vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(28) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(29) The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(30) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(31) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(32) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(33) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g.,

the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(34) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(35) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(36) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(37) FIG. 2 is a diagram illustrating an example of a wireless communication system according to various embodiments. Referring to FIG. 2, as a part of nodes using wireless channels in a wireless communication system, a terminal **210** and a base station **220** are illustrated. The terminal **210** may also be connected to multiple base stations. Although not illustrated in FIG. 1, the base stations may be connected to the terminal **210** through multiple connectivity (e.g., dual connectivity (DC)).

(38) The terminal **210** may access a data network via the base station **220** and a user plane function (UPF). In order to transmit or receive data via the UPF, the terminal is required to generate a packet data unit (PDU) session, and one PDU session may include one or more quality of service (QoS) flows. The terminal **210** may be referred to as “a user equipment (UE)”, “a mobile station”, “a subscriber station”, “a remote terminal”, “a wireless terminal”, “a user device”, “a customer-premises equipment (CPE)”, “an electronic device”, or another term having a technical meaning equivalent thereto. The terminal **210** may correspond to the electronic device **101** illustrated in FIG. 1.

(39) The base station **220** is a network infrastructure that provides wireless access to the terminal **210** as a radio access network (RAN) node. The base station **220** has a coverage defined as a predetermined (e.g., specified) geographic area, based on a signal-transmittable distance. The base station **220** may cover one cell or cover multiple cells. The multiple cells may be distinguished based on at least one of an area of a covered sector or a supported frequency. The base station **220** may be referred to, for example, other than “a base station”, as “an access point (AP)”, “an eNodeB (eNB)”, “a 5th generation (5G) node”, “a 5G NodeB”, “a next generation node B (gNB)”, “a wireless point”, “a transmission/reception point (TRP)”, “a distributed unit (DU)”, “a radio unit (RU)”, “a remote radio headroom (RRH)”, or other terms having a technical meaning equivalent thereto.

(40) At the time of network access, the base station **220** may allocate a maximum resource to the terminal **210**. For example, the base station may allocate, to the terminal, a maximum available network resource (e.g., the number of component carriers (CCs), the number of mmWave chip transmit elements, the number of multiple-input multiple-output (MIMO) layers, and a bandwidth in an operating band). Therefore, in order to control power consumption and heating, the terminal **210** uses a scheme of sequentially controlling or interrupting functions required for communication, based on a resulting situation (e.g., overheating, or low voltage of a battery).

(41) However, in this scheme, long term evolution (LTE) fallback control due to heat from the terminal **210** may be applied. In addition, due to the application of LTE fallback control, the terminal may fail to sufficiently secure a 5.sup.th generation (5G) connection maintenance time. Moreover, since it is impossible to apply LTE fallback control to the terminal in a 5G standalone (SA) situation, there is a problem in that heating control has to be performed in a 5G connection state having high power consumption.

(42) Hereinafter, various example embodiments of the disclosure provide an apparatus and a method for securing a 5G connection maintenance time and controlling heating by being allocated by the terminal **210** for a basic resource required for 5G communication from the base station **220** at the time of network access, and requesting an additional resource, based on data throughput.

(43) FIG. 3 is a signal flow diagram illustrating an example operation of signaling for resource allocation according to various embodiments. With reference to FIG. 3, a procedure in which the terminal **210** optimizes power consumption through signaling with the base station **220**, and prevents and/or reduces overheating is described.

(44) Referring to FIG. 3, in operation **310**, the terminal may transmit user equipment (UE) capability information to the base station **220**. The UE capability information may correspond to a request related to a basic resource. The basic resource may be related, for example, and without limitation, to at least one of the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band. For example, the basic resource may be one CC, single-input single-output (SISO), and a minimum bandwidth (e.g., 100 MHz) in an operating band (e.g., n261). Although not illustrated in FIG. 3, in some embodiments, the base station **220** may transmit a UE capability enquiry message to the terminal **210**. The terminal **210** may transmit the UE capability information to the base station **220** in response to the UE capability enquiry message received from the base station **220**.

(45) According to an embodiment, the UE capability information may include information

indicating a basic resource request. For example, the information indicating the basic resource request may be indicated by one bit.

(46) According to an embodiment, the UE capability information may include information on a basic resource. The information on the basic resource may include at least one of information on the number of CCs, the number of MIMO layers, or a bandwidth in an operating band. For example, the terminal **210** may include, in the capability information, an information element (IE) configured to have “1” as the number of CCs supported thereby, and transmit the information. For example, the terminal **210** may include, in the capability information, an IE configured to have “1” as the number of MIMO layers supported thereby, and transmit the information. For example, the terminal **210** may include, in the capability information, an IE configured to have “100 MHz” as a bandwidth within an operating band, which is supported thereby, and transmit the information.

(47) According to an embodiment, when a basic resource is requested, the UE capability information may not include information on a maximum resource supported by the terminal **210**. That is, when a basic resource is requested, so as to reduce the overhead of the UE capability information, the UE capability information may not include information on a maximum resource supported by the terminal **210**. The terminal **210** may configure, as a value related to a basic resource requested by the terminal **210**, a value corresponding to each information element (IE) of the UE capability information.

(48) In operation **320**, the base station **220** may allocate a basic resource to the terminal **210**. According to an embodiment, the base station **220** may allocate the basic resource to the terminal **210**, based on the received UE capability information. According to an embodiment, the base station **220** may indicate information related to the allocated basic resource to the terminal **210**, based, for example, and without limitation, on at least one of a radio resource control (RRC) message, a medium access control control element (MAC CE), and downlink control information (DCI). For example, the allocation of the basic resource to the terminal **210** at the time of network access may reduce or prevent the overheating of the terminal **210**.

(49) In operation **330**, the terminal **210** may transmit an additional resource request message to the base station **220**. According to an embodiment, the terminal **210** may transmit the additional resource request message to the base station **220**, based on data throughput. When the data throughput exceeds a threshold value (or it is expected that same exceeds the threshold value), the terminal **210** may transmit the additional resource request message to the base station **220**. The threshold value may include one or more values. For example, the terminal **210** may configure one or more stages of transmitting an additional resource request message through one or more values. For example, when the data throughput of the terminal **210** to which the basic resource has been allocated exceeds a first threshold value, the terminal **210** may transmit the additional resource request message to the base station **220**. When the data throughput of the terminal **210** to which a resource has been additionally allocated through the additional resource request message exceeds a second threshold value, the terminal may transmit the additional resource request message to the base station **220**. The second threshold value is greater than the first threshold value. The threshold value may be called an additional resource request trigger value.

(50) According to an embodiment, the additional resource request message may include information on an additional resource. The information on the additional resource may be identified based on a temperature margin of the terminal **210**. The temperature margin may refer, for example, to the difference between the temperature measured through a thermistor inside the terminal **210**, and a temperature controlling or interrupting a function of the terminal **210**. The information on the additional resource may include at least one of information on the number of CCs, information on a bandwidth, or information on the number of MIMO layers. The information on the additional resource may be configured based on an operation situation (e.g., an application being in use) of the terminal **210**. The information on the number of CCs may include at least one of information on the number of downlink CCs, or information on the number of uplink CCs. The

information on the bandwidth may include at least one of information on a downlink bandwidth within frequency range (FR) 1, information on an uplink bandwidth within FR1, information on a downlink bandwidth within FR2, or information on an uplink bandwidth within FR2. The information on the number of MIMO layers may include at least one of information on the number of downlink MIMO layers within FR1, information on the number of uplink MIMO layers within FR1, information on the number of downlink MIMO layers within FR2, or information on the number of uplink MIMO layers within FR2. The base station **220** may allocate the additional resource to the terminal **210**, based on the information on the additional resource.

(51) According to an embodiment, the additional resource request message may correspond to an overheating assistance message. The information on the additional resource may be indicated to the base station **220** through one or more parameters defined in the overheating assistance message. For example, the one or more parameters related to the additional resource may include at least one of a parameter related to the number of downlink CCs (e.g., reducedCCsDL), a parameter related to the number of uplink CCs (e.g., reducedCCsUL), a parameter related to a downlink bandwidth within frequency range (FR) 1 (e.g., reducedBW-FR1-DL), a parameter related to an uplink bandwidth within FR1 (e.g., reducedBW-FR1-UL), a parameter related to a downlink bandwidth within FR2 (e.g., reducedBW-FR2-DL), a parameter related to an uplink bandwidth within FR2 (e.g., reducedBW-FR2-UL), a parameter related to the number of downlink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-DL), a parameter related to the number of uplink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-UL), a parameter related to the number of downlink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-DL), and a parameter related to the number of uplink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-UL).

(52) According to an embodiment, the additional resource request message may correspond to a tracking area update (TAU) request message. TAU may be a procedure of updating a tracking area (TA) to which the terminal **210** belongs, in order to manage the mobility of the terminal **210**. The terminal **210** may periodically transmit a TAU request message to the base station **220**. When data throughput exceeds a threshold value, the TAU request message may include information requesting update of UE capability information. The information requesting update of the UE capability information may be configured as a UE radio capability information update needed field in the TAU request message. Whether update of the UE capability information is needed may be indicated by one bit. The UE radio capability information update needed field may be configured as shown in Table 1 and Table 2 below.

(53) TABLE-US-00001 TABLE 1 8 7 6 5 4 3 2 1 UE radio capability 0 0 0 URC octet 1
information spare upd update needed IEI

(54) TABLE-US-00002 TABLE 2 UE radio capability information update needed flag (URC upd)
(octet 1) Bit 1 0 UE radio capability information update not needed 1 UE radio capability
information update needed

(55) Although not illustrated in FIG. 3, the base station **220** may transmit a UE capability enquiry message to the terminal **210**, based on the received TAU request message. The terminal **210** may transmit UE capability information to the base station **220** in response to the UE capability enquiry message. The transmitted UE capability information may include at least one of information on the number of CCs, information on the number of MIMO layers, information on a bandwidth, information on a band, information on a modulation order, information on a scaling factor, or information on a subcarrier spacing (SCS), which are supported by the terminal **210**. The information on the number of CCs may include at least one of information on the number of maximum CCs supported by the terminal **210** in downlink, or information on the number of maximum CCs supported by the terminal **210** in uplink. The information on the number of MIMO layers may include at least one of information on the number of maximum layers supported by the terminal **210** in downlink reception, or information on the number of maximum layers supported by the terminal **210** in uplink transmission. The information on the bandwidth may include

information on a channel bandwidth supported by the terminal **210** for an SCS. The information on the band may include information on a frequency band number supported by the terminal **210**. The information on the modulation order may include at least one of information on a maximum modulation order supported by the terminal **210** in downlink, or information on a maximum modulation order supported by the terminal **210** in uplink. The information on the scaling factor may include a factor to be applied to a band in calculating a maximum data rate of the terminal **210**. The information on the SCS may include at least one of information on an SCS supported by the terminal **210** in downlink, or information on an SCS supported by the terminal **210** in uplink. The base station **220** may allocate an additional resource to the terminal **210**, based on the received UE capability information.

(56) In FIG. **3**, signaling requesting additional resource allocation from the base station **220** according to data throughput of the terminal **210** after the basic resource is allocated is illustrated. According to embodiments of the disclosure, after a maximum resource is allocated, a basic resource may be allocated to the terminal **210** from the base station **220** through signaling. Hereinafter, an embodiment of being allocated a basic resource from the base station **220** through a user equipment (UE) assistance information message is described in greater detail below with reference to FIG. **4**.

(57) FIG. **4** is a signal flow diagram illustrating an example operation of signaling for resource allocation according to various embodiments. With reference to FIG. **4**, a procedure in which the terminal **210** optimizes power consumption through signaling with the base station **220**, and prevents and/or reduces overheating is described.

(58) Referring to FIG. **4**, in operation **410**, the terminal **210** may transmit user equipment (UE) capability information to the base station **220**. The terminal **210** may transmit the UE capability information to the base station **220** in response to a UE capability enquiry message received from the base station **220**. According to an embodiment, the UE capability information may include at least one of information on the number of CCs, information on the number of MIMO layers, information on a bandwidth, information on a band, information on a modulation order, information on a scaling factor, or information on a subcarrier spacing (SCS), which are supported by the terminal **210**. The information on the number of CCs may include at least one of information on the number of maximum CCs supported by the terminal **210** in downlink, or information on the number of maximum CCs supported by the terminal **210** in uplink. The information on the number of MIMO layers may include at least one of information on the number of maximum layers supported by the terminal **210** in downlink reception, or information on the number of maximum layers supported by the terminal **210** in uplink transmission. The information on the bandwidth may include information on a channel bandwidth supported by the terminal **210** for an SCS. The information on the band may include information on a frequency band number supported by the terminal **210**. The information on the modulation order may include at least one of information on a maximum modulation order supported by the terminal **210** in downlink, or information on a maximum modulation order supported by the terminal **210** in uplink. The information on the scaling factor may include a factor to be applied to a band in calculating a maximum data rate of the terminal **210**. The information on the SCS may include at least one of information on an SCS supported by the terminal **210** in downlink, or information on an SCS supported by the terminal **210** in uplink. The base station **220** may allocate a resource to the terminal **210**, based on the received UE capability information.

(59) In operation **420**, the terminal **210** may transmit an overheating assistance message to the base station **220**. The overheating assistance message may include one or more parameter related to a basic resource. The one or more parameters related to the basic resource may include at least one of a parameter related to the number of downlink CCs (e.g., reducedCCsDL), a parameter related to the number of uplink CCs (e.g., reducedCCsUL), a parameter related to a downlink bandwidth within frequency range (FR) 1 (e.g., reducedBW-FR1-DL), a parameter related to an uplink

bandwidth within FR1 (e.g., reducedBW-FR1-UL), a parameter related to a downlink bandwidth within FR2 (e.g., reducedBW-FR2-DL), a parameter related to an uplink bandwidth within FR2 (e.g., reducedBW-FR2-UL), a parameter related to the number of downlink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-DL), a parameter related to the number of uplink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-UL), a parameter related to the number of downlink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-DL), and a parameter related to the number of uplink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-UL). For example, the one or more parameters included in the overheating assistance message may be configured as shown in Table 3 below. According to an embodiment, when internal overheating is detected, the terminal **210** may transmit the overheating assistance message to the base station **220**. According to an embodiment, when it is expected that internal overheating will not occur, the terminal **210** may transmit the overheating assistance message to the base station **220**. For example, when internal overheating has not occurred for a predetermined period, the terminal **210** may transmit the overheating assistance message to the base station **220**. According to an embodiment, the overheating assistance message may be included in a UE assistance information message and then be transmitted.

(60) TABLE-US-00003 TABLE 3 UEAssistanceInformation-v1540-IEs ::= SEQUENCE {
overheatingAssistance OverheatingAssistance OPTIONAL, nonCriticalExtension
UEAssistanceInformation- v16xy-IEs OPTIONAL } OverheatingAssistance ::= SEQUENCE
{ reducedMaxCCs SEQUENCE { reducedCCsDL INTEGER (0..31),
reducedCCsUL INTEGER (0..31) } OPTIONAL, reducedMaxBW-FR1
SEQUENCE { reducedBW-FR1-DL ReducedAggregatedBandwidth,
reducedBW-FR1-UL ReducedAggregatedBandwidth } OPTIONAL, reducedMaxBW-FR2
SEQUENCE { reducedBW-FR2-DL ReducedAggregatedBandwidth,
reducedBW-FR2-UL ReducedAggregatedBandwidth } OPTIONAL, reducedMaxMIMO-
LayersFR1 SEQUENCE { reducedMIMO-LayersFR1-DL MIMO-
LayersDL, reducedMIMO-LayersFR1-UL MIMO-LayersUL } OPTIONAL,
reducedMaxMIMO-LayersFR2 SEQUENCE { reducedMIMO-LayersFR2-DL
MIMO-LayersDL, reducedMIMO-LayersFR2-UL MIMO-LayersUL
} OPTIONAL }

(61) In operation **430**, the base station **220** may allocate a basic resource to the terminal **210**. According to an embodiment, the base station **220** may allocate the basic resource to the terminal **210**, based on the received overheating assistance message. For example, the terminal **210** may limit the resource allocated through the overheating assistance message to the basic resource. According to an embodiment, the base station **220** may indicate information related to the allocated basic resource to the terminal, based on at least one of a radio resource control (RRC) message, a medium access control control element (MAC CE), and downlink control information (DCI). The allocation of the basic resource to the terminal **210** may reduce or prevent the overheating of the terminal **210**. In operation **440**, the terminal **210** may transmit an additional resource request message to the base station **220**. The operation **440** in FIG. 4 corresponds to the operation **330** in FIG. 3, and thus a detailed description thereof may not be repeated here.

(62) FIG. 5 is a signal flow diagram illustrating an example operation of signaling for identifying the number of antenna elements of a terminal according to various embodiments. With reference to FIG. 5, a procedure of identifying the number of used antenna elements of an antenna array of the terminal **210** (specifically, a fixed-type device) is described.

(63) Referring to Table 4 below, the number of antenna elements of a 5th generation (5G) patch antenna array is proportional to the power consumption of the terminal **210**.

(64) TABLE-US-00004 TABLE 4 The number of Tx 55° C. 85° C. antenna elements TRP (dBm)
power (mW) power (mW) 1 0 148.7 149.7 4 6 374.7 378.7 8 9 701.2 709.2 1 14 364.7 369.7 4 20
1239.7 1257.7 8 23 2432.2 2455.2

(65) For example, when the terminal **210** is a fixed-type device (e.g., a customer-premises equipment (CPE)), the terminal **210** and the base station **220** has a constant distance therebetween. Therefore, so as to minimize and/or reduce unnecessary power consumption and an increase in device temperature, the number of minimum antenna elements to be used by the terminal **210** for communication is required to be identified. Referring to FIG. 5, in operation **510**, the terminal **210** may receive a reference signal from the base station **220**. According to an embodiment, the reference signal may include a synchronization signal. The synchronization signal may be a synchronization signal and physical broadcast channel (SS/PBCH) block. According to an embodiment, the reference signal may include a cell specific reference signal (CRS), a channel state information-reference signal (CSI-RS), a channel state information-interference measurement (CSI-IM), or a combination thereof.

(66) In operation **520**, the terminal **210** may identify the number of minimum antenna elements. The terminal **210** may identify a received signal strength, based on the received reference signal. The terminal **210** may identify the number of minimum antenna elements, based on the identified received signal strength. The received signal strength may be one of a reference signal receive quality (RSRQ), a reference signal received power (RSRP) or a reference signal strength indicator (RSSI). The number of minimum antenna elements may be the number of antenna elements corresponding to a power for maintaining a signal through beamforming with the base station **220**.

(67) In operation **530**, the terminal **210** may transmit received signal strength information to the base station **220**. The received signal strength information may include at least one of information on an RSRQ, an RSRP, an RSSI, or the number of minimum antenna elements.

(68) According to an embodiment, the operations based on FIG. 5 may be performed before radio resource control (RRC) connection. The terminal **210** or the base station **220** may also perform operations of various embodiments according to the disclosure after performing the operations based on FIG. 5.

(69) FIG. 6 is a flowchart illustrating an example operation of a terminal for configuring information on an additional resource according to various embodiments.

(70) Referring to FIG. 6, in operation **610**, a basic resource (e.g., a bandwidth of 100 MHz, single-input single-output (SISO), and one component carrier (CC)) may be allocated to the terminal **210** from the bandwidth. According to an embodiment, the terminal **210** may transmit UE capability information corresponding to a basic resource request to the base station **220**. The base station **220** may allocate a basic resource to the terminal **210** in response to the reception of the UE capability information corresponding to the basic resource request. According to an embodiment, the terminal **210** may transmit UE capability information corresponding to a maximum resource request to the base station **220**. The base station **220** may allocate a maximum resource to the terminal **210** in response to the reception of the UE capability information corresponding to the maximum resource request. The terminal **210** may transmit an overheating assistance message to the base station **220**. The base station **220** may allocate a basic resource to the terminal **210**, based on basic resource information included in the overheating assistance message.

(71) In operation **620**, the terminal **210** may identify whether an additional resource is needed. According to an embodiment, the terminal **210** may identify whether the additional resource is needed, based on data throughput. When the data throughput exceeds a threshold value (or it is expected that same exceeds the threshold value), the terminal **210** may identify that the additional resource is needed. The threshold value may include one or more values. That is, the terminal **210** may configure one or more stages of identifying whether an additional resource is needed, through one or more values. For example, when the data throughput of the terminal **210** exceeds a first threshold value, the terminal **210** may identify that the additional resource is needed. In addition, after the additional resource is allocated, when the data throughput of the terminal **210** exceeds a second threshold value, the terminal **210** may identify that an additional resource is needed. The second threshold value is greater than the first threshold value. The threshold value may be called

an additional resource request trigger value.

(72) The terminal **210** may perform operation **630** when it is identified that the additional resource is needed (Yes in operation **620**). The terminal **210** may perform operation **620** again when it is not identified that the additional resource is needed (No in operation **620**). According to an embodiment, the terminal **210** may monitor an internal temperature for determination in operation **620**. For example, the terminal **210** may identify a temperature margin in order to identify whether an overheating assistance message is transmitted.

(73) In operation **630**, the terminal **210** may identify whether the number of allocated multiple-input multiple-output (MIMO) layers is maximum. The number of MIMO layers may include at least one of the number of downlink MIMO layers within FR1, the number of uplink MIMO layers within FR1, the number of downlink MIMO layers within FR2, or the number of uplink MIMO layers within FR2.

(74) In operation **640**, when the number of MIMO layers is not maximum (No in operation **630**), the terminal **210** may increase the number of MIMO layers. Information on the increased number of MIMO layers may be included in information on an additional resource. According to an embodiment, the terminal **210** may transmit the information on the additional resource to the base station **220**. After operation **640**, the terminal **210** may perform monitoring to identify whether an additional resource is needed again.

(75) In operation **650**, when the number of MIMO layers is maximum (Yes in operation **630**), the terminal **210** may identify whether the number of allocated component carriers (CCs) is maximum. The number of CCs may include at least one of the number of downlink CCs, or the number of uplink CCs.

(76) In operation **660**, when the number of CCs is not maximum (No in operation **650**), the terminal **210** may increase the number of CCs. Information on the increased number of CCs may be included in information on an additional resource. According to an embodiment, the terminal **210** may transmit the information on the additional resource to the base station **220**. After operation **660**, the terminal **210** may perform monitoring to identify whether an additional resource is needed again.

(77) In operation **670**, when the number of CCs is maximum (Yes in operation **650**), the terminal **210** may identify whether an allocated bandwidth is maximum. The bandwidth may include at least one of a downlink bandwidth within 1, an uplink bandwidth within FR1, a downlink bandwidth within FR2, or an uplink bandwidth within FR2.

(78) In operation **680**, when the bandwidth is not maximum (No in operation **670**), the terminal **210** may increase the bandwidth. Information on the increased bandwidth may be included in information on an additional resource. According to an embodiment, the terminal **210** may transmit the information on the additional resource to the base station **220**. After operation **680**, the terminal **210** may perform monitoring to identify whether an additional resource is needed again.

(79) When the bandwidth is maximum, the terminal **210** may terminate the operations based on FIG. **6**, and maintain a current state. According to an embodiment, the terminal **210** may perform an operation of preventing and/or reducing overheating through a function control.

(80) In FIG. **6**, an additional resource is requested in the order of the number of MIMOs, the number of CCs, and a bandwidth, but this merely corresponds to an example. According to an embodiment, when an additional resource is needed, the operations based on FIG. **6** may also be performed while putting a priority on a resource having a small increase in power consumption. Power consumption may vary according to a characteristic of the terminal, and thus a priority may vary according to a characteristic of the terminal. In FIG. **6**, an example of increasing one resource among the number of MIMOs, the number of CCs, and a bandwidth is illustrated, but this also merely corresponds to an example. According to an embodiment, the terminal **210** may transmit, to the base station **220**, additional resource information in which at least one of the number of MIMOs, the number of CCs, and a bandwidth is increased.

(81) In FIG. 6, an example of requesting an additional resource for all the number of MIMOs, the number of CCs, and a bandwidth is illustrated, but this merely corresponds to an example.

According to an embodiment, at least one of the number of MIMOs, the number of CCs, and a bandwidth may be omitted in the requested additional resource.

(82) FIG. 7 is a flowchart illustrating an example operation of a base station for resource allocation according to various embodiments. With reference to FIG. 7, an operation flow of a base station for optimizing power consumption of the terminal **210** and preventing and/or reducing overheating is described.

(83) Referring to FIG. 7, in operation **710**, the base station **220** may allocate a basic resource to the terminal **210**. According to an embodiment, the base station **220** may indicate information related to the allocated basic resource to the terminal, based on at least one of a radio resource control (RRC) message, a medium access control control element (MAC CE), or downlink control information (DCI).

(84) According to an embodiment, the base station **220** may transmit a user equipment (UE) capability enquiry message to the terminal **210**. The base station **220** may receive UE capability information from the terminal **210**. The UE capability information may correspond to a request related to a basic resource. The basic resource may be related to at least one of the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band. For example, the basic resource may be one CC, SISO, and a minimum bandwidth (e.g., 100 MHz) in an operating band (e.g., n261). According to an embodiment, the basic resource may be defined in advance. According to an embodiment, the UE capability information may include information indicating a minimum resource request. For example, the information indicating the minimum resource request may be indicated by one bit. According to an embodiment, the UE capability information may not include information on a maximum resource supported by the terminal **210**. The base station **220** may allocate a basic resource to the terminal **210**, based on the received UE capability information.

(85) According to an embodiment, the base station **220** may receive UE capability information from the terminal **210**. The UE capability information may include at least one of information on the number of CCs, information on the number of MIMO layers, information on a bandwidth, information on a band, information on a modulation order, information on a scaling factor, or information on a subcarrier spacing (SCS), which are supported by the terminal **210**. The information on the number of CCs may include at least one of information on the number of maximum CCs supported by the terminal **210** in downlink, or information on the number of maximum CCs supported by the terminal **210** in uplink. The information on the number of MIMO layers may include at least one of information on the number of maximum layers supported by the terminal **210** in downlink reception, or information on the number of maximum layers supported by the terminal **210** in uplink transmission. The information on the bandwidth may include information on a channel bandwidth supported by the terminal **210** for an SCS. The information on the band may include information on a frequency band number supported by the terminal **210**. The information on the modulation order may include at least one of information on a maximum modulation order supported by the terminal **210** in downlink, or information on a maximum modulation order supported by the terminal **210** in uplink. The information on the scaling factor may include a factor to be applied to a band in calculating a maximum data rate of the terminal **210**. The information on the SCS may include at least one of information on an SCS supported by the terminal **210** in downlink, or information on an SCS supported by the terminal **210** in uplink. The base station **220** may allocate a resource to the terminal **210**, based on the received UE capability information. The base station **220** may receive an overheating assistance message from the terminal **210**. The overheating assistance message may include one or more parameter related to a basic resource. The one or more parameters related to the basic resource may include at least one of a parameter related to the number of downlink CCs (e.g., reducedCCsDL), a parameter

related to the number of uplink CCs (e.g., reducedCCsUL), a parameter related to a downlink bandwidth within frequency range (FR) 1 (e.g., reducedBW-FR1-DL), a parameter related to an uplink bandwidth within FR1 (e.g., reducedBW-FR1-UL), a parameter related to a downlink bandwidth within FR2 (e.g., reducedBW-FR2-DL), a parameter related to an uplink bandwidth within FR2 (e.g., reducedBW-FR2-UL), a parameter related to the number of downlink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-DL), a parameter related to the number of uplink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-UL), a parameter related to the number of downlink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-DL), and a parameter related to the number of uplink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-UL). For example, the basic resource may be one CC, SISO, and a minimum bandwidth (e.g., 100 MHz) in an operating band (e.g., n261). The base station **220** may allocate a basic resource to the terminal **210**, based on received information on the basic resource.

(86) In operation **720**, the base station **220** may receive an additional resource request message from the terminal **210**.

(87) According to an embodiment, the additional resource request message may include information on an additional resource. The information on the additional resource may include at least one of information on the number of CCs, information on a bandwidth, or information on the number of MIMO layers. The information on the additional resource may be configured based on an operation situation (e.g., an application being in use) of the terminal **210**. The information on the number of CCs may include at least one of information on the number of downlink CCs, or information on the number of uplink CCs. The information on the bandwidth may include at least one of information on a downlink bandwidth within frequency range (FR) 1, information on an uplink bandwidth within FR1, information on a downlink bandwidth within FR2, or information on an uplink bandwidth within FR2. The information on the number of MIMO layers may include at least one of information on the number of downlink MIMO layers within FR1, information on the number of uplink MIMO layers within FR1, information on the number of downlink MIMO layers within FR2, and information on the number of uplink MIMO layers within FR2. According to an embodiment, the additional resource request message may correspond to an overheating assistance message.

(88) According to an embodiment, the additional resource request message may correspond to a tracking area update (TAU) request message. TAU may be a procedure of updating a tracking area (TA) in order to manage the mobility of the terminal **210**. The TAU request message may include information requesting update of UE capability information. The information requesting update of the UE capability information may be configured as a UE radio capability information update needed field in the TAU request message. Whether update of the UE capability information is needed may be indicated by one bit. The base station **220** may transmit a UE capability enquiry message to the terminal **210**, based on the received TAU request message. The base station **220** may receive UE capability information from the terminal **210**. The UE capability information may include at least one of information on the number of CCs, information on the number of MIMO layers, information on a bandwidth, information on a band, information on a modulation order, information on a scaling factor, or information on a subcarrier spacing (SCS), which are supported by the terminal **210**. The information on the number of CCs may include at least one of information on the number of maximum CCs supported by the terminal **210** in downlink, or information on the number of maximum CCs supported by the terminal **210** in uplink. The information on the number of MIMO layers may include at least one of information on the number of maximum layers supported by the terminal **210** in downlink reception, or information on the number of maximum layers supported by the terminal **210** in uplink transmission. The information on the bandwidth may include information on a channel bandwidth supported by the terminal **210** for an SCS. The information on the band may include information on a frequency band number supported by the terminal **210**. The information on the modulation order may include at least one of information on a

maximum modulation order supported by the terminal **210** in downlink, or information on a maximum modulation order supported by the terminal **210** in uplink. The information on the scaling factor may include a factor to be applied to a band in calculating a maximum data rate of the terminal **210**. The information on the SCS may include at least one of information on an SCS supported by the terminal **210** in downlink, or information on an SCS supported by the terminal **210** in uplink.

(89) In operation **730**, the base station **220** may allocate an additional resource to the terminal **210**. According to an embodiment, the base station **220** may allocate the additional resource to the terminal **210**, based on the received additional resource request message. According to an embodiment, the base station **220** may allocate the additional resource to the terminal **210**, based on the received UE capability information. According to an embodiment, the base station **220** may allocate the additional resource to the terminal **210**, based on the received overheating assistance message.

(90) FIG. **8** is a flowchart illustrating an example operation of a terminal for resource allocation according to various embodiments. With reference to FIG. **8**, an operation flow of the terminal **210** for optimizing power consumption and preventing and/or reducing overheating is described.

(91) In operation **810**, the terminal **210** may transmit UE capability information to the base station **220**.

(92) According to an embodiment, the UE capability information may correspond to a request related to a basic resource. The basic resource may be related to at least one of the number of component carriers (CC), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band. For example, the basic resource may be one CC, SISO, and a minimum bandwidth (e.g., 100 MHz) in an operating band (e.g., n261). According to an embodiment, the UE capability information may include information indicating a minimum resource request. For example, the information indicating the minimum resource request may be indicated by one bit. According to an embodiment, when the UE capability information corresponds to a request related to a basic resource, the UE capability information may not include information on a maximum resource supported by the terminal **210**. The base station **220** may allocate a basic resource to the terminal **210**, based on the received UE capability information.

(93) According to an embodiment, the UE capability information may include at least one of information on the number of CCs, information on the number of MIMO layers, information on a bandwidth, information on a band, information on a modulation order, information on a scaling factor, or information on a subcarrier spacing (SCS), which are supported by the terminal **210**. The information on the number of CCs may include at least one of information on the number of maximum CCs supported by the terminal **210** in downlink, or information on the number of maximum CCs supported by the terminal **210** in uplink. The information on the number of MIMO layers may include at least one of information on the number of maximum layers supported by the terminal **210** in downlink reception, or information on the number of maximum layers supported by the terminal **210** in uplink transmission. The information on the bandwidth may include information on a channel bandwidth supported by the terminal **210** for an SCS. The information on the band may include information on a frequency band number supported by the terminal **210**. The information on the modulation order may include at least one of information on a maximum modulation order supported by the terminal **210** in downlink, or information on a maximum modulation order supported by the terminal **210** in uplink. The information on the scaling factor may include a factor to be applied to a band in calculating a maximum data rate of the terminal **210**. The information on the SCS may include at least one of information on an SCS supported by the terminal **210** in downlink, or information on an SCS supported by the terminal **210** in uplink. The base station **220** may allocate a resource to the terminal **210**, based on the received UE capability information. The terminal **210** may transmit an overheating assistance message to the base station **220**. The overheating assistance message may include information on a basic resource.

The overheating assistance message may include one or more parameter related to the basic resource. The one or more parameters related to the basic resource may include at least one of a parameter related to the number of downlink CCs (e.g., reducedCCsDL), a parameter related to the number of uplink CCs (e.g., reducedCCsUL), a parameter related to a downlink bandwidth within frequency range (FR) 1 (e.g., reducedBW-FR1-DL), a parameter related to an uplink bandwidth within FR1 (e.g., reducedBW-FR1-UL), a parameter related to a downlink bandwidth within FR2 (e.g., reducedBW-FR2-DL), a parameter related to an uplink bandwidth within FR2 (e.g., reducedBW-FR2-UL), a parameter related to the number of downlink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-DL), a parameter related to the number of uplink MIMO layers within FR1 (e.g., reducedMIMO-LayersFR1-UL), a parameter related to the number of downlink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-DL), and a parameter related to the number of uplink MIMO layers within FR2 (e.g., reducedMIMO-LayersFR2-UL). The basic resource may be related to at least one of the number of CCs, the number of MIMO layers, or a bandwidth in an operating band. The base station **220** may allocate the basic resource to the terminal **210**, based on the received overheating assistance message.

(94) In operation **820**, the terminal **210** may identify whether an additional resource is needed. According to an embodiment, the terminal **210** may identify whether the additional resource is needed, based on data throughput. When the data throughput exceeds a threshold value (or it is expected that same exceeds the threshold value), the terminal **210** may identify that the additional resource is needed. The threshold value may include one or more values. That is, the terminal **210** may configure one or more stages enabling identifying of whether an additional resource is needed, through one or more values. For example, when the data throughput of the terminal **210** to which the basic resource has been allocated exceeds a first threshold value, the terminal **210** may identify that the additional resource is needed. In addition, after the additional resource is allocated, when the data throughput of the terminal **210** exceeds a second threshold value, the terminal **210** may identify that an additional resource is needed. The second threshold value is greater than the first threshold value. The threshold value may be called an additional resource request trigger value.

(95) In operation **830**, the terminal **210** may transmit an additional resource request message to the base station **220**. When it is identified that the additional resource is needed, the terminal **210** may transmit an additional resource request message to the base station **220**.

(96) According to an embodiment, an additional resource allocation request message may include information on an additional resource. The information on the additional resource may be identified based on a temperature margin of the terminal **210**. The temperature margin may be the difference between the temperature measured through a thermistor inside the terminal **210**, and a temperature controlling or interrupting a function of the terminal **210**. The information on the additional resource may include at least one of information on the number of CCs, information on a bandwidth, or information on the number of MIMO layers. The information on the number of CCs may include at least one of information on the number of downlink CCs, or information on the number of uplink CCs. The information on the bandwidth may include at least one of information on a downlink bandwidth within frequency range (FR) 1, information on an uplink bandwidth within FR1, information on a downlink bandwidth within FR2, or information on an uplink bandwidth within FR2. The information on the number of MIMO layers may include at least one of information on the number of downlink MIMO layers within FR1, information on the number of uplink MIMO layers within FR1, information on the number of downlink MIMO layers within FR2, or information on the number of uplink MIMO layers within FR2. The base station **220** may allocate an additional resource to the terminal **210**, based on the information on the additional resource. According to an embodiment, the additional resource request message may be called an overheating assistance message.

(97) According to an embodiment, the additional resource request message may correspond to a tracking area update (TAU) request message. TAU may be a procedure of updating a tracking area

(TA) to which the terminal **210** belongs, in order to manage the mobility of the terminal **210**. The terminal **210** may periodically transmit a TAU request message to the base station **220**. When data throughput exceeds a threshold value, the TAU request message may include information requesting update of UE capability information. The information requesting update of the UE capability information may be configured as a UE radio capability information update needed field in the TAU request message. Whether update of the UE capability information is needed may be indicated by one bit. The base station **220** may transmit a UE capability enquiry message to the terminal **210**, based on the received TAU request message. The terminal **210** may transmit UE capability information to the base station **220** in response to the UE capability enquiry message. The UE capability information may include at least one of information on the number of CCs, information on the number of MIMO layers, information on a bandwidth, information on a band, information on a modulation order, information on a scaling factor, or information on a subcarrier spacing (SCS), which are supported by the terminal **210**. The information on the number of CCs may include at least one of information on the number of maximum CCs supported by the terminal **210** in downlink, or information on the number of maximum CCs supported by the terminal **210** in uplink. The information on the number of MIMO layers may include at least one of information on the number of maximum layers supported by the terminal **210** in downlink reception, or information on the number of maximum layers supported by the terminal **210** in uplink transmission. The information on the bandwidth may include information on a channel bandwidth supported by the terminal **210** for an SCS. The information on the band may include information on a frequency band number supported by the terminal **210**. The information on the modulation order may include at least one of information on a maximum modulation order supported by the terminal **210** in downlink, or information on a maximum modulation order supported by the terminal **210** in uplink. The information on the scaling factor may include a factor to be applied to a band in calculating a maximum data rate of the terminal **210**. The information on the SCS may include at least one of information on an SCS supported by the terminal **210** in downlink, or information on an SCS supported by the terminal **210** in uplink. The base station **220** may allocate an additional resource to the terminal **210**, based on the received UE capability information.

(98) Through FIGS. **1, 2, 3, 4, 5, 6, 7** and FIG. **8**, initiatively/pre-emptively controlling resource allocation by a terminal has been described. In a conventional 5th generation (5G) communication system, when a terminal accesses a network, a maximum resource is allocated to the terminal, and thus power may be excessively consumed, and long term evolution (LTE) due to overheating may occur. However, a general scenario (e.g., Internet browsing, a three-dimensional (3D) game, or watching high-quality video) does not require a high data throughput, and thus 5G communication may be possible only with basic resource allocation. For example, referring to Table 5 and Table 6 below, while a recommended bit rate for a high dynamic range (HDR) upload is only a maximum of 85 Mbps, data throughput based on a basic resource runs to 323.2 Mbps.

(99) TABLE-US-00005

TABLE 5 Recommended bit rate for standard dynamic range (SDR) upload											
High Standard frame Recommended bit rate for high frame rate rate, dynamic range (HDR) upload (SFR), (HFR) SFR, HFR, Type Mbps Mbps Type Mbps Mbps											
2160p (4k)	35-45	53-68	2160p (4k)	44-56	66-85	1440p (2k)	16	24	1440p (2k)	20	30
1080p	8	12	1080p	10	15	720p	5	7.5	720p	6.5	9.5
480p	2.5	4	480p	Not	Not	supported	supported	360p	1	1.5	360p
Not	Not	supported	supported	360p	1	1.5	360p	Not	Not	supported	supported

(100) TABLE-US-00006

TABLE 6 FR2 throughput											
Bandwidth	100 MHz	Subcarrier spacing (SCS)	120 kHz	Modulation	64 QAM	Multiple-input multiple-output (MIMO)	Single-input single-output (SISO)	Component carrier (CC)	1 DL:UL	Ratio (4:1)	0.8
Resource block (RB)	66	Subcarrier	12	Orthogonal frequency division	56 multiplexing (OFDM)	symbol	Data bit	6	Layer (component carrier (CC) or MIMO)	1	Code rate
0.92578125	Overhead	0.18	TP, bps	404,032,860	TP, Mbps	404.0	Downlink TP, Mbps	323.2			

(101) Therefore, according to various embodiments of the disclosure, only a basic network

resource may be allocated to a terminal to perform 5G communication, and a network resource may be additionally allocated thereto by a determination and request of the terminal itself, and thus optimization of power consumption and overheating prevention and/or reduction may be possible. (102) An electronic device according to an example embodiment of the disclosure may include: a communication circuit and at least one processor operatively connected to the communication circuit, wherein the at least one processor is configured to: control the communication circuit to transmit first capability information of the electronic device to a base station; identify, based on data throughput, whether an additional resource is needed; and based on identifying that the additional resource is needed, control the communication circuit to transmit an additional resource request message to the base station, the first capability information corresponding to a request related to a basic resource, and the basic resource being related to at least one of the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band.

(103) According to an example embodiment, the additional resource request message may include at least one of information on the number of downlink CCs, information on the number of uplink CCs, information on the number of downlink MIMO layers, information on the number of uplink MIMO layers, information on a downlink bandwidth, or information on an uplink bandwidth.

(104) According to an example embodiment, the at least one processor may be configured to: control the communication circuit to receive a user equipment (UE) capability enquiry message from the base station, and transmit second capability information of the electronic device to the base station, the additional resource request message may correspond to a tracking area update (TAU) request message, and the TAU request message may include a request to update the first capability information.

(105) According to an example embodiment, the at least one processor may be configured to: control the communication circuit to receive a reference signal from the base station; and identify, based on a received strength of the reference signal, a minimum number of antenna elements.

(106) According to an example embodiment, the at least one processor may be configured to: based on the data throughput exceeding a threshold value, control the communication circuit to transmit the additional resource request message to the base station; and identify, based on an internal temperature margin of the electronic device, information included in the additional resource request message, wherein the internal temperature margin may correspond to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

(107) According to an example embodiment, the electronic device may correspond to a customer-premises equipment (CPE).

(108) An electronic device according to an example embodiment of the disclosure may include: a communication circuit and at least one processor operatively connected to the communication circuit, wherein the at least one processor is configured to: control the communication circuit to transmit first capability information of the electronic device to a base station and transmit an overheating assistance message to the base station; identify, based on data throughput, whether an additional resource is needed; and based on identifying that the additional resource is needed, control the communication circuit to transmit an additional resource request message to the base station, the overheating assistance message including information on a basic resource, and the information on the basic resource including at least one of information on the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth.

(109) According to an example embodiment, the additional resource request message may include at least one of information on the number of downlink CCs, information on the number of uplink CCs, information on the number of downlink MIMO layers, information on the number of uplink MIMO layers, information on a downlink bandwidth, or information on an uplink bandwidth.

(110) According to an example embodiment, the at least one processor may be configured to: control the communication circuit to receive a user equipment (UE) capability enquiry message from the base station, and transmit second capability information of the electronic device to the base station, the additional resource request message may correspond to a tracking area update (TAU) request message, and the TAU request message may include a request to update the first capability information.

(111) According to an example embodiment, the at least one processor may be configured to: control the communication circuit to receive a reference signal from the base station; and identify, based on a received strength of the reference signal, a minimum number of antenna elements.

(112) According to an example embodiment, the at least one processor may be configured to: based on the data throughput exceeding a threshold value, control the communication circuit to transmit the additional resource request message to the base station; and identify, based on an internal temperature margin of the electronic device, information included in the additional resource request message, wherein the internal temperature margin may correspond to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

(113) According to an example embodiment, the electronic device may correspond to a customer-premises equipment (CPE).

(114) A method performed by an electronic device according to an example embodiment of the disclosure may include: transmitting first capability information of the electronic device to a base station, identifying, based on data throughput, whether an additional resource is needed, and based on identifying that the additional resource is needed, transmitting an additional resource request message to the base station, wherein the first capability information corresponds to a request related to a basic resource, and the basic resource is related to at least one of the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth in an operating band.

(115) According to an example embodiment, the additional resource request message may include at least one of information on the number of downlink CCs, information on the number of uplink CCs, information on the number of downlink MIMO layers, information on the number of uplink MIMO layers, information on a downlink bandwidth, or information on an uplink bandwidth.

(116) According to an example embodiment, the method may include: receiving a user equipment (UE) capability enquiry message from the base station, and transmitting second capability information of the electronic device to the base station, wherein the additional resource request message corresponds to a tracking area update (TAU) request message, and the TAU request message includes a request to update the first capability information.

(117) According to an example embodiment, the method may include receiving a reference signal from the base station, and identifying, based on a received strength of the reference signal, a minimum number of antenna elements.

(118) According to an example embodiment, the method may include: identifying, based on an internal temperature margin of the electronic device, information included in the additional resource request message, wherein the additional resource request message is transmitted based on the data throughput exceeding a threshold value, wherein the internal temperature margin corresponds to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

(119) According to an example embodiment, the electronic device may correspond to a customer-premises equipment (CPE).

(120) A method performed by an electronic device according to an example embodiment of the disclosure may include: transmitting first capability information of the electronic device to a base station, transmitting an overheating assistance message to the base station, identifying, based on data throughput, whether an additional resource is needed, and based on identifying that the

additional resource is needed, transmitting an additional resource request message to the base station, wherein the overheating assistance message includes information on a basic resource, and the information on the basic resource includes at least one of information on the number of component carriers (CCs), the number of multiple-input multiple-output (MIMO) layers, or a bandwidth.

(121) According to an example embodiment, the additional resource request message may include at least one of information on the number of downlink CCs, information on the number of uplink CCs, information on the number of downlink MIMO layers, information on the number of uplink MIMO layers, information on a downlink bandwidth, or information on an uplink bandwidth.

(122) According to an example embodiment, the method may include receiving a user equipment (UE) capability enquiry message from the base station, and transmitting second capability information of the electronic device to the base station, wherein the additional resource request message corresponds to a tracking area update (TAU) request message, and the TAU request message includes a request to update the first capability information.

(123) According to an example embodiment, the method may include receiving a reference signal from the base station, and identifying, based on a received strength of the reference signal, the number of minimum antenna elements.

(124) According to an example embodiment, the method may include identifying, based on an internal temperature margin of the electronic device, information included in the additional resource request message, wherein the additional resource request message is transmitted based on the data throughput exceeding a threshold value, and the internal temperature margin corresponds to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

(125) According to an example embodiment, the electronic device may correspond to a customer-premises equipment (CPE).

(126) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(127) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(128) As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof,

adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(129) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(130) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(131) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(132) While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various modification may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device in a wireless communication system, the electronic device comprising: a communication circuit; at least one processor, comprising processing circuitry, electrically connected to the communication circuit; and memory storing instructions that, when executed by the at least one processor individually or collectively, cause the electronic device to: transmit, to a

base station, first capability information of the electronic device using the communication circuit, wherein the first capability information includes first information for requesting first resources to be assigned for controlling an internal temperature of the electronic device, receive, from the base station, information indicating that the first resources for controlling the internal temperature are assigned to the electronic device, in case that data throughput is greater than a threshold configured for triggering a request of additional resources, identify that second resources corresponding to the additional resources are needed for an additional data transmission, and based on identifying that the second resources are needed, transmit, to the base station, a tracking area update (TAU) message for requesting the second resources, wherein the TAU message includes second information indicating that update of the first capability information is needed for requesting the second resources, wherein the first resources correspond to minimum and/or low resources for a communication between the electronic device and the base station, wherein the first information includes at least one of a number of component carriers (CCs) for controlling the internal temperature, a number of multiple-input multiple-output (MIMO) layers for controlling the internal temperature, or a bandwidth for controlling the internal temperature, and wherein the second information includes at least one of a maximum and/or large number of CCs supported by the electronic device, a maximum and/or large number of MIMO layers supported by the electronic device, or a maximum bandwidth supported by the electronic device.

2. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: receive, from the base station, a reference signal using the communication circuit, and identify, based on a strength of the received reference signal, a minimum number of antenna elements.

3. The electronic device of claim 1, wherein information on an internal temperature margin of the electronic device is included in the TAU message for requesting the second resources, and wherein the internal temperature margin corresponds to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

4. The electronic device of claim 2, wherein the electronic device comprises a customer-premises equipment (CPE).

5. An electronic device in a wireless communication system, the electronic device comprising: a communication circuit; and at least one processor, comprising processing circuitry, electrically connected to the communication circuit, and configured to: transmit, to a base station, first capability information of the electronic device using the communication circuit, wherein the first capability information includes first information for requesting first resources to be assigned for controlling an internal temperature of the electronic device, receive, from the base station, information indicating that the first resources for controlling the internal temperature are assigned to the electronic device, in case that data throughput is greater than a threshold configured for triggering a request of additional resources, identify that second resources corresponding to the additional resources are needed for an additional data transmission, and based on identifying that the second resources are needed, transmit, to the base station, a tracking area update (TAU) message for requesting the second resources, wherein the TAU message includes second information indicating that update of the first capability information is needed for requesting the second resources, wherein the first resources correspond to minimum resources for a communication between the electronic device and the base station, wherein the first information includes at least one of a number of component carriers (CCs) for controlling the internal temperature, a number of multiple-input multiple-output (MIMO) layers for controlling the internal temperature, or a bandwidth for controlling the internal temperature, and wherein the second information includes at least one of a maximum number of CCs supported by the electronic device, a maximum number of MIMO layers supported by the electronic device, or a maximum bandwidth supported by the electronic device.

6. The electronic device of claim 5, wherein the at least one processor is configured to: receive,

from the base station, a reference signal using the communication circuit, and identify, based on a strength of the received reference signal, a minimum number of antenna elements.

7. The electronic device of claim 5, wherein information on an internal temperature margin of the electronic device is included in the TAU message for requesting the second resources, and wherein the internal temperature margin corresponds to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

8. The electronic device of claim 6, wherein the electronic device comprises a customer-premises equipment (CPE).

9. A method performed by an electronic device in a wireless communication system, the method comprising: transmitting, to a base station, first capability information of the electronic device using a communication circuit, wherein the first capability information includes first information for requesting first resources to be assigned for controlling an internal temperature of the electronic device; receiving, from the base station, information indicating that the first resources for controlling the internal temperature are assigned to the electronic device; in case that data throughput is greater than a threshold configured for triggering a request of additional resources, identifying that second resources corresponding to the additional resources are needed for an additional data transmission; and based on identifying that the second resources are needed, transmitting, to the base station, a tracking area update (TAU) message for requesting the second resources, wherein the TAU message includes second information indicating that update of the first capability information is needed for requesting the second resources, wherein the first resources correspond to minimum resources for a communication between the electronic device and the base station, wherein the first information includes at least one of a number of component carriers (CCs) for controlling the internal temperature, a number of multiple-input multiple-output (MIMO) layers for controlling the internal temperature, or a bandwidth for controlling the internal temperature, and wherein the second information includes at least one of a maximum number of CCs supported by the electronic device, a maximum number of MIMO layers supported by the electronic device, or a maximum bandwidth supported by the electronic device.

10. The method of claim 9, further comprising: receiving, from the base station, a reference signal; and identifying, based on a strength of the received reference signal, a minimum number of antenna elements.

11. The method of claim 9, wherein information on an internal temperature margin of the electronic device is included in the TAU message for requesting the second resources, and wherein the internal temperature margin corresponds to a difference between a measured temperature value of the electronic device and a value specified for a function control of the electronic device.

12. The method of claim 10, wherein the electronic device comprises a customer-premises equipment (CPE).
